

EDU GENIE

AI-Powered Learning Assistant

1. Project Overview

EduGenie is an AI-powered educational assistant that helps students learn more effectively by combining several study tools into a single platform, powered by Google Gemini.

Instead of switching between different apps to ask questions, understand new concepts, test themselves, summarize material, and plan what to study next, students can do all of it through one assistant.

2. Problem Statement

Students today face a few common challenges:

- Struggling to find quick, simple explanations for difficult concepts
- Spending too much time creating self-test questions to check understanding
- Difficulty summarizing long study material efficiently
- Lack of a clear, structured path for learning a new topic from scratch

EduGenie addresses all four problems through one AI-driven assistant.

3. Key Features

Question & Answer

Ask any academic question and receive a clear, concise, easy-to-understand answer — with step-by-step explanations when needed.

Concept Explanation

Get any concept broken down into a simple, beginner-friendly structure: Meaning, Easy Explanation, Real-world Example, and Key Points.

Quiz Generator

Automatically generates a 3-question multiple-choice quiz on any topic, helping students test their understanding instantly.

Text Summarizer

Converts long notes or study material into a short, clear, bullet-point summary — without losing important information.

Personalized Learning Path

Creates a complete, structured study plan for any topic and skill level, including learning goals, prerequisites, a step-by-step path, practice activities, mini projects, and expected outcomes.

4. How It Works

1. The student submits a question, topic, or text through the web interface.
2. The request is sent to the EduGenie backend.
3. The backend builds a specialized instruction for Google Gemini, tailored to the type of request.
4. Gemini generates the response.
5. EduGenie formats and returns the result to the student in a clean, easy-to-read format.

5. Technology Used

Layer	Technology
Backend Framework	FastAPI (Python)
AI Engine	Google Gemini
Frontend Templates	Jinja2 (HTML)
Server	Uvicorn

6. Target Users

- College and university students
- Self-learners exploring new subjects
- Students preparing for exams who need quick revision and self-testing tools

7. Benefits

- All-in-one — combines 5 essential study tools in a single assistant
- Simple language — every response is tailored for easy, college-level understanding
- Time-saving — reduces hours spent creating notes, quizzes, and study plans manually
- Structured learning — encourages a step-by-step approach rather than random studying
- Scalable — built on a modular backend that can easily support more features in the future

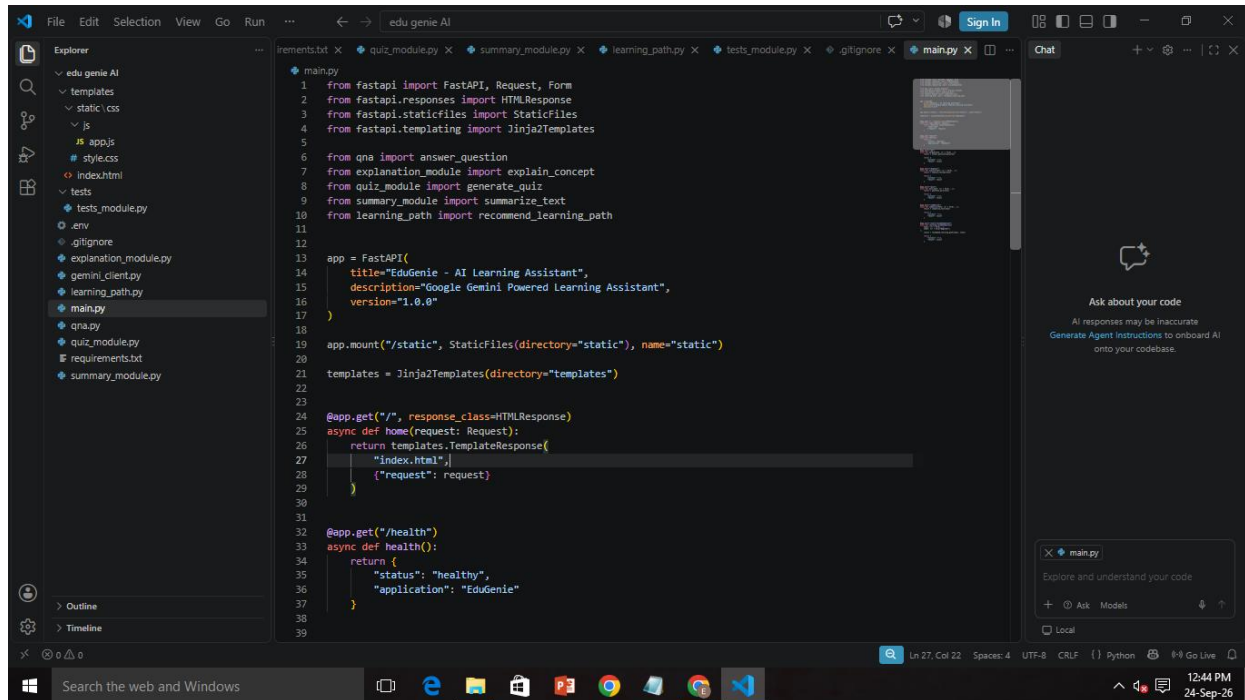
8. Future Enhancements

- User accounts and progress tracking
- Saving quiz scores and learning history
- Support for file uploads (PDFs, notes) as input
- Voice-based question input
- Multi-language support

9. Code Implementation Snapshots

The following screenshots show the core backend implementation of EduGenie in Visual Studio Code.

9.1 main.py — FastAPI Application & Routing

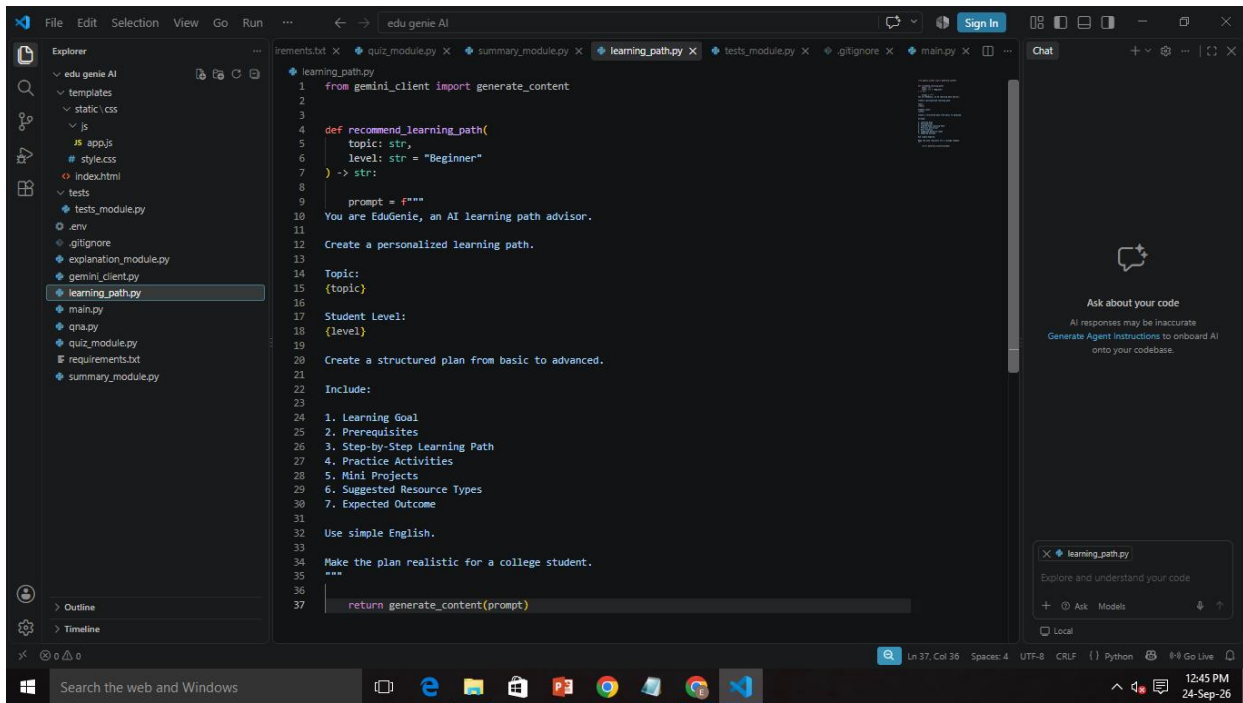


```
1 from fastapi import FastAPI, Request, Form
2 from fastapi.responses import HTMLResponse
3 from fastapi.staticfiles import StaticFiles
4 from fastapi.templating import Jinja2Templates
5
6 from qna import answer_question
7 from explanation_module import explain_concept
8 from quiz_module import generate_quiz
9 from summary_module import summarize_text
10 from learning_path import recommend_learning_path
11
12
13 app = FastAPI(
14     title="EduGenie - AI Learning Assistant",
15     description="Google Gemini Powered Learning Assistant",
16     version="1.0.0"
17 )
18
19 app.mount("/static", StaticFiles(directory="static"), name="static")
20
21 templates = Jinja2Templates(directory="templates")
22
23
24 @app.get("/", response_class=HTMLResponse)
25 async def home(request: Request):
26     return templates.TemplateResponse(
27         "index.html",
28         {"request": request}
29     )
30
31
32 @app.get("/health")
33 async def health():
34     return {
35         "status": "healthy",
36         "application": "EduGenie"
37     }
38
39
```

FastAPI app setup, static/template mounting, and the home ("/") and health-check routes.

- Acts as the entry point of the entire application, wiring together all five AI feature modules.
- Registers clean, RESTful routes (/qa, /explain, /quiz, /summarize, /learn/recommendations) for easy frontend integration.
- Includes a dedicated /health endpoint, making the app easy to monitor in production.
- Uses Jinja2Templates to serve a dynamic home page alongside the API.

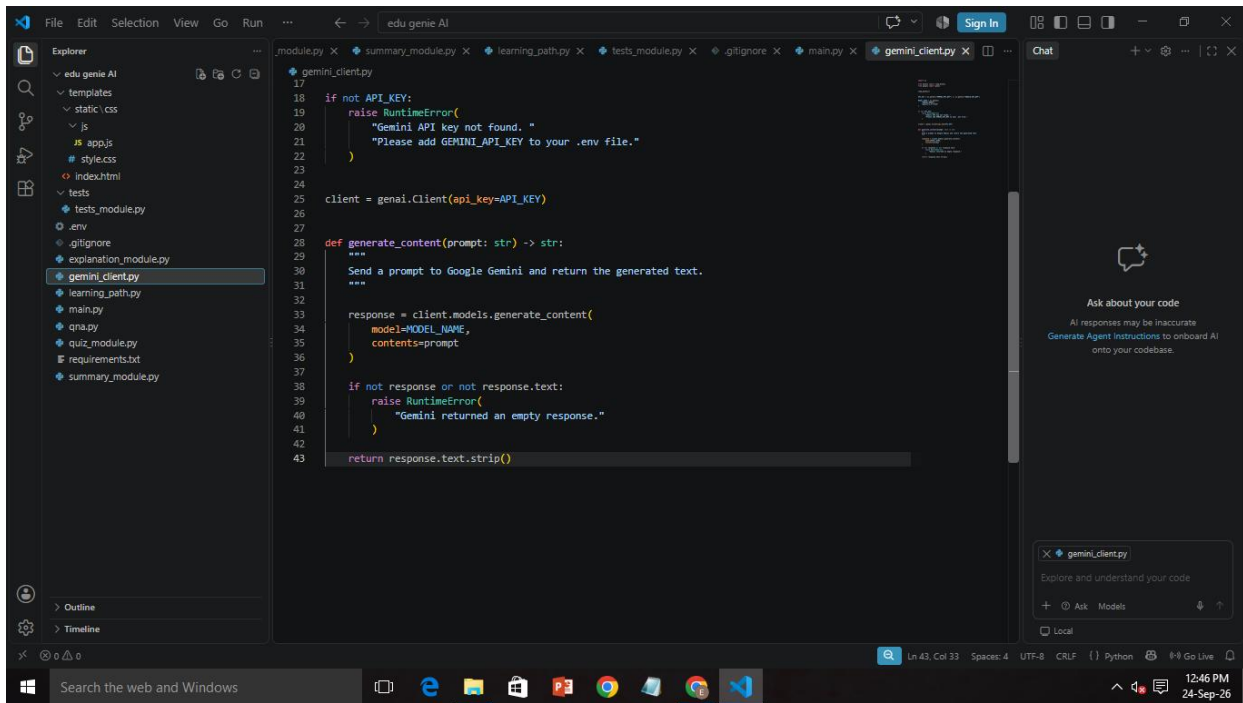
9.2 learning_path.py — Personalized Learning Path Module



Builds the prompt for Google Gemini to generate a structured, level-based learning path.

- Turns a single topic into a complete 7-part roadmap — from goals and prerequisites to mini-projects.
- Accepts a student level (Beginner by default), so the same function adapts to different learners.
- Demonstrates prompt engineering in action: clear role, clear structure, clear constraints.
- Reuses the shared generate_content() connector, keeping the module short and focused.

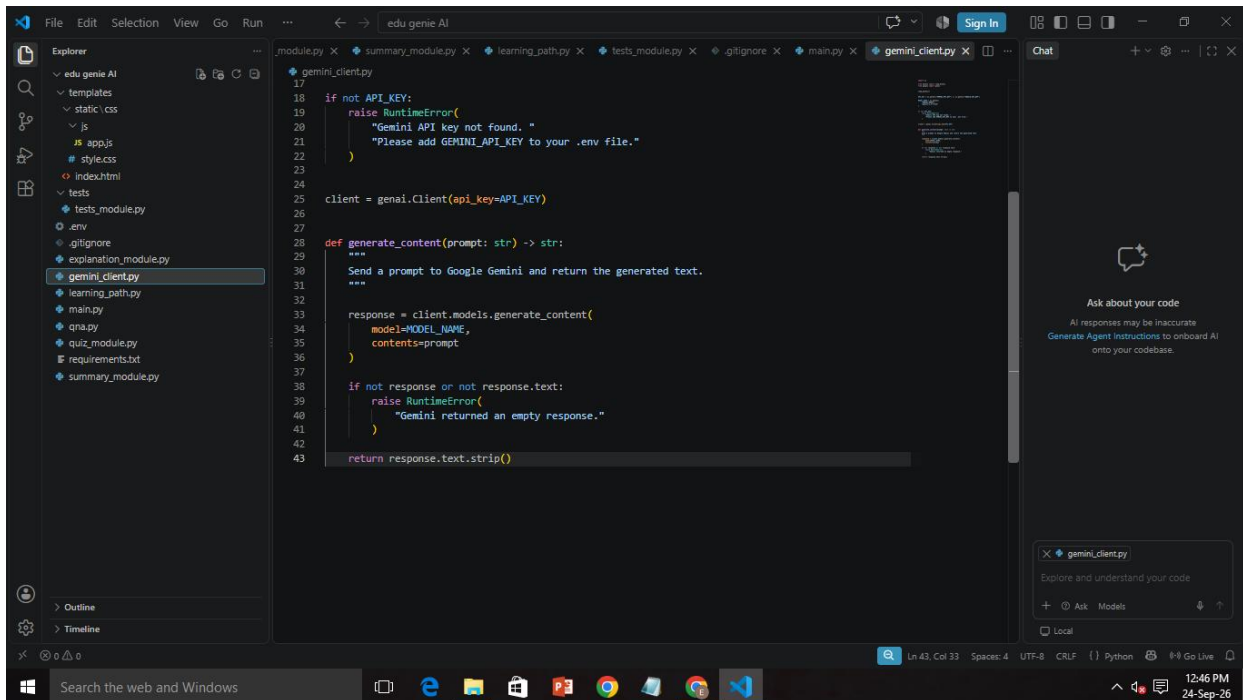
9.3 gemini_client.py — Google Gemini Client Setup (Part 1)



Initializes the Gemini API client and defines `generate_content()`, the core function used by every module.

- The backbone of the entire project — every AI feature ultimately calls this one function.
- Fails fast with a clear error if the API key is missing, avoiding confusing bugs later.
- Centralizing the Gemini connection here means the AI provider could be swapped with minimal changes elsewhere.

9.4 gemini_client.py — Google Gemini Client Setup (Part 2)



Same core connector file — the single function all feature modules call to talk to Gemini.

- Guards against empty Gemini responses, so downstream modules never silently process blank text.
- Returns clean, trimmed text — one less thing every feature module needs to handle itself.
- A small file with outsized importance: reliable, reusable, and easy to test in isolation.

10. Conclusion

EduGenie brings together the power of Google Gemini and a clean, easy-to-use interface to give students a single, reliable AI companion for learning — from asking questions to building a full study plan.